

CSE 2106: Microprocessor and Assembly Language Lab (Batch 30) ----- Lab Evaluation 3

[Solve Two Problems Based on Your roll%3+1]

1. Write a Cortex-M4 assembly language program that calls a function **remove_duplicates** that removes duplicate elements from a sorted array and returns new length.
2. Write a Cortex-M4 assembly language program that calls a function **Print_Prime_Number** to generate all the prime numbers up to N and store them in an array.
3. Write a Cortex-M4 assembly language program that calls a function **Find_GCD** to find the Greatest Common Divisor (GCD) of two numbers (R0, R1) using Euclid's algorithm.
4. Write a Cortex-M4 assembly language program that calls a function **Fibonacci_Sequence**, to generate the first N Fibonacci numbers and store them in an array.
5. Write a Cortex-M4 assembly program that calls a function **string_compare** that compares two strings lexicographically.

Input: R0 = string1, R1 = string2 **Output:** R2 = 0 (equal), negative (str1 < str2), positive (str1 > str2)

6. Write a Cortex-M4 assembly program that calls a function **count_character** which counts occurrences of a specific character in a null-terminated string.

Input: R0 = string address, R1 = character to count **Output:** R2 = count of character occurrences